

Lab 7 Fillable Worksheet

Integrated Malware Analysis

Use this worksheet while completing Lab 7. Record your findings from static analysis, debugging, system monitoring, persistence analysis, and network traffic analysis. When finished, choose **Print / Save as PDF** to submit your completed worksheet.

Safety reminder: Analyze malware only inside an isolated virtual machine or a controlled sandbox such as **ANY.RUN**. Never run malware on your host system.

Student Information

Student name

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Date

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Course / section

CYB-310-002

Instructor

Troy Adams

Sample Identification

Malware sample name

Remcos RAT JavaScript dropper

Source repository

MalwareBazaar

File name

RFQ-R632090.JS

File size

6'480'135 bytes

SHA-256 hash

bfaf3900078db99c433f5d6e1d58989ae2c7c5a81aabeebc4668a87a89790466

Known or suspected malware family

RemcosRAT.

Environment Preparation

Preparation item	Done	Notes
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Preparation item	Done	Notes
Created a clean VM snapshot before analysis	Yes	
Configured isolated or controlled networking	Yes	VM was setup to be isolated. ANY.RUN was used to show networking execution.
Disabled shared folders and unsafe host integration	Yes	
Started monitoring tools before execution	Yes	Used ProcMon and Wireshark.
Recorded baseline system conditions	Yes	

Part 1 — Static Analysis

Static analysis task	Findings
File hashes and identification	SHA-256:bfaf3900078db99c433f5d6e1d58989ae2c7c5a81aabeebc4668a87a89790466 SHA1: a8527c290eca33866358f967c43d0e1594498eec MD5: 75d02379ba86ef84b2dbcac00c856a07 File Type: js File Name: RFQ-R632090.JS
PE headers or section observations	The initial sample is a JavaScript file, so both PE studio and Detect It Easy do not apply to the original artifact. A dropped PE executable (wscript.exe -> OOXFVYBGLAVGHMXI.exe) is shown during runtime behavior but that reconstructed payload was not successfully recovered so manual PE inspection was unsuccessful.
Imports and suspicious APIs	Manual inspection of the JavaScript revealed suspicious scripting and file write behavior, including references to WScript and strings such as SaveToFile and CopyFile.
Strings, URLs, IPs, paths, or mutexes	Also heavy obfuscated strings and repeated payload building concatenation were observed. Later runtime analysis ties the sample to mynewraw.duckdns.org, IP 107.175.249.142 and the dropped file OOXFVYBGLAVGHMXI.exe and browser related artifacts such as Login Data.tmp and Login Data For Account.tmp
Packing, obfuscation, or protections detected	The entire script is very heavily obfuscated. Showing randomized variable names, indexed decoder calls such as functionName(0x1fa) and repeated string concatenation used to build a large payload buffer. Which ties into MalwareBazaar also labeling the sample as obfuscated, packed, evasive and anti-debug related.

Before execution, what did static analysis suggest the malware might do?

Static inspection suggested that the JavaScript file functions as a dropper. This is clearly indicated after it is ran but with the heavy obfuscation, decoder logic, WScript usage and SaveToFile/CopyFile strings it strongly indicates without having to execute that it reconstructs a second-stage payload and then writes it to disk before executing it. This behavior is consistent with a staged Remcos delivery chain.

Part 2 — Dynamic Execution and Debugging

Entry point address

No valid PE entry point

Debugger used

x64dbg was attempted

Suspicious API or behavior	Why it matters	Observed runtime evidence
WScript.exe execution	Confirms script-based malware execution through Windows Script Host.	The sample launches through wscript.exe in the observed process chain.
Dropped executable launched from suspicious path	Indicates staged payload delivery rather than a standalone script-only threat.	Runtime behavior shows wscript.exe dropping and starting C:\Users\Public\...\OOXFVYBGLAVGHMXI.exe.
File-write behavior consistent with SaveToFile / COM object abuse	Shows the script is likely reconstructing and writing a second-stage payload to disk.	Manual script inspection revealed SaveToFile and CopyFile, while Windows Scripting Host queries a suspicious COM object and executes a dropped EXE.
Outbound network communication associated with Remcos RAT	Supports remote access trojan behavior and command-and-control capability.	App.Any.Run revealed mynewraw.duckdns.org and IP 107.175.249.142.

Breakpoints that triggered

Debugger-based breakpoint analysis was not completed because no valid PE was available for full x64dbg analysis.

Processes spawned, injected into, or terminated

Observed execution chain: winrar.exe → wscript.exe → OOXFVYBGLAVGHMXI.exe → conhost.exe.

Memory allocation, unpacking, injection, or evasion findings

Not manually confirmed in the debugger. However, MalwareBazaar/App.Any.Run reports for this sample flags debugger-detection behavior and suspicious WriteProcessMemory, so evasive and stealth-oriented behavior is likely even though it was not directly proven in local debugger work.

Part 3 — System and Persistence Analysis

Artifact type	Observed evidence
Files created, dropped, or modified	Runtime evidence shows the script dropped C:\Users\Public\...\OOXFVYBGLAVGHMXI.exe and C:\Users\Public\...\OOXFVYBGLAVGHMXI.PNG. The Remcos payload also dropped browser-related SQLite artifacts including Login Data.tmp and Login Data For Account.tmp.
Registry changes	ANY.RUN recorded heavy registry interaction, including 2,970 total registry events, 2,930 reads, 27 writes, and 13 deletes. Local ProcMon also showed WScript.exe performing many RegOpenKey and RegQueryValue operations across HKLM and HKCU, including paths related to OLE, WMI security, Image File Execution Options, display settings, and other Windows configuration areas.
New or injected processes	New processes observed during execution include wscript.exe, the randomized dropped executable OOXFVYBGLAVGHMXI.exe, and conhost.exe.
Scheduled tasks or services	No scheduled task or service creation was conclusively observed.
Run / RunOnce keys or startup entries	No Run or RunOnce persistence entry was conclusively observed in the collected evidence.

Did the malware establish persistence? If yes, explain how

Persistence was not conclusively proven from the current evidence set. The malware clearly performed significant registry activity, but a specific startup mechanism such as Run key, service, or scheduled task was not directly confirmed.

Part 4 — Network Traffic Analysis

Network observation	Details
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DNS queries	Local Wireshark capture primarily showed multicast DNS and local service discovery traffic, including repeated PTR queries for _ocular_sp._tcp.local and related variants. In contrast, sandbox intelligence showed external DNS activity associated with mynewraw.duckdns.org.
Domains contacted	The strongest malicious domain observed was mynewraw.duckdns.org. MalwareBazaar-linked behavior also notes use of dynamic DNS services.
IP addresses contacted	The key malicious IP associated with the dropped payload was 107.175.249.142.
Protocols observed	Local Wireshark mainly showed UDP/mDNS activity and DHCPv6. The ANY.RUN report recorded broader runtime communication including DNS, HTTP, and TCP/IP traffic.
Downloaded payloads, beaconing, or command-and-control behavior	The script acted as a dropper that executed a second-stage EXE, and the dropped Remcos payload was associated with external network communication through dynamic DNS infrastructure. That combination is consistent with remote access trojan command-and-control behavior.

Part 5 — Correlation of Findings

Correlation question	Response
Which imports or strings seen statically were confirmed during execution?	Static inspection revealed obfuscated scripting behavior, a custom decoder, WScript, and SaveToFile/CopyFile. These findings were confirmed dynamically by the observed wscript.exe execution chain and the dropped executable launched at runtime.
Which registry paths, files, or startup locations were hinted at statically and later observed dynamically?	Static inspection suggested that the script would reconstruct and write files to disk. Dynamic analysis confirmed dropped files in C:\Users\Public\... and additional browser-related SQLite artifacts. Startup persistence was not conclusively confirmed.
Did expected network behavior match the traffic you actually captured?	Partially. Based on the Remcos/dropper logic, external communication was expected. Local Wireshark mainly captured mDNS/local discovery traffic, but the sandbox clearly showed broader DNS, HTTP, and TCP/IP communication tied to the malware's runtime behavior.
What evidence from multiple tools best explains what the malware does?	Manual static analysis showed an obfuscated JavaScript dropper that decodes strings, builds a payload buffer, and references SaveToFile. Dynamic analysis then showed wscript.exe dropping and launching a randomized executable identified as Remcos RAT, along with registry interaction, dropped browser-data files, and external network communication. Together, these findings show a staged JavaScript dropper delivering a Remcos remote access trojan payload.

Indicators of Compromise (IOCs)

Indicator type	Value
File hash	bfaf3900078db99c433f5d6e1d58989ae2c7c5a81aabeebc4668a87a89790466

Indicator type	Value
Domain	mynewraw.duckdns.org
IP address	107.175.249.142
Registry key	Not conclusively identified as a stable IOC
Dropped file	C:\Users\Public...\OOXFVYBGLAVGHMXI.exe
Service, task, mutex, or startup entry	Not conclusively identified

Behavioral Summary and Reflection

Provide a consolidated summary of the malware's behavior

The analyzed sample is a heavily obfuscated JavaScript dropper. Manual inspection showed decoder obfuscation, randomized variable names, repeated payload building concatenation and file/write indicators such as SaveToFile and CopyFile. Dynamic analysis then showed execution through wscript.exe, dropping and launching OOXFVYBGLAVGHMXI.exe, which was identified as Remcos RAT. The payload performed extensive registry interaction, dropped browser related SQLite files and communicated with external infrastructure associated with dynamic DNS. Overall, the evidence supports classification of the malware sample as a staged Remcos infection chain.

What indicators or artifacts could defenders use for detection and response?

Useful detection artifacts include the SHA-256 hash, the domain mynewraw.duckdns.org, the IP 107.175.249.142, the dropped executable path under C:\Users\Public\..., execution through wscript.exe and suspicious script behavior involving COM/file/write methods such as SaveToFile.

How did combining multiple tools improve your understanding of the sample?

Using multiple tools made the malware's full behavior much clearer. Manual script inspection explained how the obfuscation and file-write logic worked, ProcMon showed that WScript.exe actively queried the registry and interacted with the system and ANY.RUN provided the overall process tree, dropped files, malware family identification and external network indicators.

Did ANY.RUN or another sandbox reveal behavior you did not initially observe manually?

Yes. ANY.RUN revealed the full staged process chain, the exact dropped executable name, the dropped PNG and SQLite artifacts, the external infrastructure tied to mynewraw.duckdns.org. Those details were not all visible from local ProcMon and Wireshark alone.

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